

STUDYHUB
CLASS 12 - MP BOARD
BIOLOGY — SET 2
English Medium | Original Sample Paper

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 2

This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions carrying a total of 28 marks.
3. Questions 6 to 12 carry 2 marks each.
4. Questions 13 to 16 carry 3 marks each.
5. Questions 17 to 20 carry 4 marks each.
6. Internal choices are provided from Question 6 to Question 20.
7. Draw neat and labelled diagrams wherever necessary.
8. Use appropriate biological terms in your answers.

SECTION - A

OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct option. $1 \times 6 = 6$

(i) Which structure in a flower receives pollen grains?

- A. Anther
- B. Stigma
- C. Ovary
- D. Filament

(ii) Which hormone stimulates milk production after childbirth?

- A. Prolactin
- B. Oxytocin
- C. FSH
- D. LH

(iii) A cross between an individual of unknown genotype and a homozygous recessive individual is called:

- A. Monohybrid cross
- B. Dihybrid cross
- C. Test cross
- D. Back mutation

(iv) Which nitrogenous base is present in RNA but normally absent in DNA?

- A. Adenine

- B. Thymine
- C. Guanine
- D. Uracil

(v) Which microorganism is commonly used in the production of alcohol?

- A. Yeast
- B. Rhizobium
- C. Lactobacillus
- D. Penicillium

(vi) The relationship in which one organism benefits and the other is neither harmed nor benefited is:

- A. Parasitism
- B. Commensalism
- C. Mutualism
- D. Predation

Q.2 Fill in the blanks. 1×6=6

- (i) The nutritive tissue formed after double fertilisation in angiosperms is called _____.
- (ii) The hormone responsible for the release of milk from mammary glands is _____.
- (iii) The physical appearance of an organism is called its _____.
- (iv) The sequence of three nucleotides on mRNA that specifies an amino acid is called a _____.
- (v) The process of introducing desired genetic material into a host cell is called _____.
- (vi) The variety of living organisms present in a particular region is called _____.

Q.3 Write True or False. 1×5=5

- (i) Cleistogamous flowers are always self-pollinated.
- (ii) Ovulation generally occurs around the middle of a normal menstrual cycle.
- (iii) A recessive trait is expressed only in the heterozygous condition.
- (iv) DNA replication is semiconservative.
- (v) Energy decreases as we move from one trophic level to the next.

Q.4 Match Column A with Column B. 1×6=6

Column A	Column B
(i) Pollen tube	(a) Translation
(ii) Corpus luteum	(b) Carries male gametes
(iii) Ribosome	(c) Secretes progesterone
(iv) Operon	(d) Gene regulation
(v) Penicillin	(e) Antibiotic
(vi) Decomposer	(f) Nutrient recycling

Q.5 Answer in one word/one sentence. 1×5=5

- (i) Name the part of the ovule through which the pollen tube usually enters.
- (ii) What is the fusion of male and female gametes called?
- (iii) What is the alternate form of a gene called?

(iv) Name the scientist who proposed the operon model.

(v) What is the first trophic level of a food chain?

SECTION - B

VERY SHORT ANSWER QUESTIONS

Q.6 2 Marks

What is apomixis? Mention one significance of apomixis in agriculture.

OR

What is polyembryony? Give one example.

Q.7 2 Marks

Differentiate between spermatogenesis and oogenesis.

OR

What is amniocentesis? Why is its misuse legally restricted?

Q.8 2 Marks

What is a dihybrid cross? State its typical F_2 phenotypic ratio.

OR

Differentiate between genotype and phenotype with one example.

Q.9 2 Marks

What is a mutation? Mention one type of mutation.

OR

What is the role of mRNA in protein synthesis?

Q.10 2 Marks

What is acquired immunity? How is it different from innate immunity?

OR

What is an antigen? How does it stimulate an immune response?

Q.11 2 Marks

What is gene cloning? State one use of gene cloning.

OR

What is a selectable marker? Why is it used in recombinant DNA technology?

Q.12 2 Marks

What is a food chain? Give one example.

OR

What is ecological pyramid? Name the three major types of ecological pyramids.

SECTION - C

SHORT ANSWER QUESTIONS

Q.13 3 Marks

Explain the process of fertilisation in flowering plants from pollen germination to the formation of the zygote.

OR

Describe the development of the embryo sac in a flowering plant.

Q.14 3 Marks

Explain a dihybrid cross with suitable symbols and state the F₂ phenotypic ratio.

OR

Explain the inheritance of haemophilia in humans as a sex-linked disorder.

Q.15 3 Marks

Explain the lac operon model of gene regulation in *E. coli*.

OR

Describe the important features of the genetic code.

Q.16 3 Marks

Explain the steps involved in the production of a recombinant DNA molecule.

OR

Describe the role of a bioreactor in biotechnology.

SECTION - D

LONG ANSWER QUESTIONS

Q.17 4 Marks

Describe the different methods of contraception used for preventing unwanted pregnancy. Mention their basic principles.

OR

Explain the process of fertilisation, implantation and early development of the human embryo.

Q.18 4 Marks

Explain DNA packaging in a eukaryotic cell. What is the role of histone proteins?

OR

Describe the process of transcription in a eukaryotic cell and explain the role of RNA polymerase.

Q.19 4 Marks

Explain the major steps involved in the production of a genetically modified organism. Mention two applications of genetically modified organisms.

OR

Explain the principle of gene therapy and describe its application in the treatment of a genetic disorder.

Q.20 4 Marks

Explain energy flow in an ecosystem. Why is the energy flow considered unidirectional?

OR

Explain the major causes of biodiversity loss and describe suitable measures for biodiversity conservation.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	5
Q.4	6
Q.5	5
Q.6-Q.12	14
Q.13-Q.16	12
Q.17-Q.20	16
TOTAL	70

$$6 + 6 + 5 + 6 + 5 + 14 + 12 + 16 = 70$$

SHORT ANSWER KEY

Q.1: B, A, C, D, A, B

Q.2: Endosperm; prolactin; phenotype; codon; transformation; biodiversity

Q.3: True, True, False, True, True

Q.4:

(i)–(b)

(ii)–(c)

(iii)–(a)

(iv)–(d)

(v)–(e)

(vi)–(f)

Q.5: Micropyle; fertilisation; allele; Jacob/François Jacob and Jacques Monod; producers

Q.8: Dihybrid F₂ phenotypic ratio = 9 : 3 : 3 : 1

Q.14: Haemophilia is an X-linked recessive disorder.

Q.15: The lac operon regulates lactose metabolism in E. coli.

Q.18: DNA is organised around histone proteins to form nucleosomes.

Q.20: Energy flow in an ecosystem is unidirectional, from producers through successive trophic levels.

Note: For Q.6–Q.20, marks should be awarded for correct biological concepts, diagrams, examples and explanations. Full solutions are not included.